

Sparse Inverse Covariance Estimation for Causal Inference in Process Data Analytics

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Abstract—Causal analysis plays a vital role in determining the underlying relationship among the variables in a system from the data. In this article, the sparse inverse covariance (SIC) estimation is coupled with likelihood score, and a two-step approach is proposed to address the problem of causal analysis. The estimation of SIC matrix for undirected sparse network reconstruction is performed with the L_0 -norm constraint in the framework of greedy sparse simplex (GSS) algorithm. Furthermore, the GSS algorithm is suitably modified to incorporate the additional constraint of positive semidefiniteness of the inverse covariance matrix. To determine the causal direction among the variables, the likelihood score is computed for the associated variables in the reconstructed network in the second step. The efficacy of the proposed approach for causal analysis is illustrated using numerical examples and an industrial application on prediction of flooding and weeping in a deethanizer column associated with a fluid catalytic cracking unit. From these studies, it is observed that the proposed approach is able to recover causal connections accurately in both cases. Furthermore, the probable reasons for the occurrence of flooding and weeping phenomena in an industrial deethanizer unit are also inferred from the identified causal network.

Index Terms—Cause effect analysis, distillation column, graphical models, maximum likelihood (ML) estimation, sparse optimization.

I. INTRODUCTION

CAUSAL inference emphasizes on decoding the relationship among the variables from the data. Based on the knowledge gained from causal analysis, applications across different domains, such as economics, reaction networks, and gene networks, are available in the literature [1], [2]. In the domain of process systems engineering, causal relationships are generally embedded in process data and process flow sheets. Using the information of causal relationship in process

systems, several challenges, such as root cause analysis of process faults [3], detection of plant-wide oscillations [4], and plant connectivity reconstruction [5], can be addressed. To understand the causal relationships, graphical models are generally considered as a viable tool and different techniques have been proposed in the literature [1], [6] to address this problem. Among these methods, time series models, such as vector autoregressive (VAR) models, have been used to determine the causality [7]. Similarly, vector autoregressive exogenous (VARX) models along with factor analysis (FA) models are used for causal analysis along with contemporaneous correlation feature inference in the variational Bayesian framework [8]. A Bayesian network structure is also learned from the data using a score-based learning algorithm [9] to determine the causal relationship. Apart from the aforementioned techniques, causal relations can also be determined using the method of intervention of variables [10]. During construction of the causal graph, it can be noted that not all variables are associated with one another. Therefore, the concept of sparsity is introduced in causal modeling in order to reduce the detection of spurious connections in a network [11]–[13].

The problem of inferring the connectivity from data can also be formulated as an estimation of inverse covariance matrix, which encodes the information of conditional independence [14], [15] among the variables. Therefore, the underlying relationships among the variables can be inferred from the data by estimating the inverse of covariance matrix in the framework of maximum likelihood (ML) estimation. In general, ML estimate of the inverse covariance matrix will lead to dense graphical model [15]. Therefore, inferring the connectivity among the variables becomes a challenging task and it may also lead to spurious connections. In order to overcome this issue, a sparsity constraint in terms of an L_1 -norm constraint on the inverse of covariance matrix is considered [16]. Different approaches, such as first-order methods [17], block coordinate ascent (BCA) [14], and alternating direction method of multipliers (ADMM) [18], are available to solve the aforementioned problem. Since inferring the causal direction is also important in the field of process systems engineering, it is essential to have methods, which can infer the directed graph from data. However, many of the aforementioned methods infer an undirected graph from the data by estimating sparse inverse covariance (SIC) matrix as a Gaussian graphical model. Given an undirected graph,

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the causal direction can be determined by computing the likelihood score, which is a metric that is used to compute the causal direction among the variables [19], [20].

In this article, a two-step algorithm is proposed for estimation of a directed graph from the data. In the first step, an undirected graph is identified by estimating the SIC matrix and the causal direction is determined through the likelihood score. Since L_0 -norm promotes better sparsity over L_1 -norm [21], a zero-norm constraint is considered in the first step while estimating the SIC matrix in order to reconstruct a sparse relational network of the process from the data. The existing likelihood score is extended for dynamic models and the likelihood score is computed in the second step to infer causal direction among the associated variables in the reconstructed sparse network from the first step. As mentioned earlier, the inverse covariance matrix has the information of Gaussian graphical model. Therefore, when sparsity constraint is introduced, some of the elements in the inverse covariance matrix become zero, which makes a variable conditionally independent of other variables. Due to the existence of conditional independence among the variables [20], information in the Gaussian graphical models can be interpreted from a causality perspective.

The main contributions of this article are summarized as follows. To solve the aforementioned nonconvex optimization problem in the first step of the proposed method, a greedy approach is used to achieve a near global optimal solution. In particular, due to its accuracy over others, the greedy sparse simplex (GSS) method [22] is considered and is adapted to the current framework of estimation. The algorithm is also modified to account for the additional positive semidefinite constraint on the inverse covariance matrix. To infer the causal direction, a likelihood score is utilized. The proposed approach for causal analysis is illustrated with numerical examples and an industrial case study. In particular, the root cause and causal map for flooding and weeping in a deethanizer associated with fluid catalytic cracking (FCC) is identified in the industrial case study.

The remainder of this article is organized as follows. In Section II, necessary preliminaries are introduced. Section III explains the details of GSS method to estimate the SIC matrix followed by an algorithmic sequence to implement the causal analysis. Section IV contains the results of the proposed method on the numerical and industrial case studies. The conclusion of the work is presented in Section V.

The following mathematical notations are used in this article. Bold uppercase letters denote a matrix, bold lowercase letters denote a vector, and the scalars are denoted by lowercase letters. Operators, $\mathbb{E}(\cdot)$, $\text{Tr}(\cdot)$, and $\|\cdot\|$, denote expectation, trace, and norm, respectively. Superscripts, $(\cdot)^{-1}$ and $(\cdot)^T$, denote inverse and transpose, respectively.

II. PRELIMINARIES

A. Graph

A graph (G) is defined as a set of objects that are related to each other in some aspect. Mathematically, a graph (G) is represented using a set of nodes (N) and edges (E) as

$$G = (N, E)$$

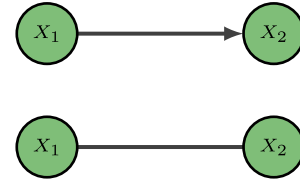


Fig. 1. Representation of directed and undirected graphs.

where N denotes the set of all variables and E denotes the set of ordered pairs where each pair represents the link connecting the variables. A graph can be classified into directed and undirected, as shown in Fig. 1. In a directed graph, nodes are connected by edges with an arrow pointed at a particular direction, whereas in an undirected graph, edges do not have the information about the direction.

A graph can also be represented in the form of a square matrix, known as adjacency matrix ($\mathbf{A}_{adj}$). In an undirected graph, adjacency matrix will be symmetric with a value of one at the (i, j) th and (j, i) th components of the matrix if the variables i and j are connected in the graph and zero otherwise. For the directed graph, only the (i, j) th component takes a value of one if the information is transferred from variable i to j and zero otherwise. A graph structure can also be constructed from the adjacency matrix by connecting the nonzero entries of the matrix. However, in practice, the adjacency matrix is not known *a priori*. In this context, methods for estimating the graph structure from data have gained significance in the recent years [23]. In Section II-B, the method of inverse covariance estimation, an approach to estimate the graph structure from the data, will be discussed.

B. SIC Estimation

In this article, the data matrix is represented by $\mathbf{X} \in \mathbb{R}^{M \times N}$, where N is the number of samples and M is the number of variables. It is further assumed that the data follow the multivariate Gaussian distribution with zero mean ($\boldsymbol{\mu} \in \mathbb{R}^{M \times 1}$) and covariance $\boldsymbol{\Sigma} \in \mathbb{R}^{M \times M}$, and then, the following proposition holds true.

Proposition 1: Given a graph G with M variables, $\mathbf{x} := \{x_i, i = 1, 2, \dots, M\}$ following a multivariate Gaussian distribution, i.e., $\mathbf{x} \sim \mathcal{N}(0, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is regular and let $\Gamma = \{1, 2, \dots, M\}$. Then, for $i, j \in \Gamma$ with $i \neq j$

$$x_i \perp\!\!\!\perp x_j | x_{\Gamma \setminus \{i, j\}} \iff (\boldsymbol{\Sigma}^{-1})_{i, j} = 0$$

i.e., the inverse of the covariance matrix encodes the information of conditional independence among the variables.

Proof: Refer to [15, Proposition 5.2]. $\square$

The conditional independence property presented in Proposition 1 also holds for multivariate Gaussian time series data [24] i.e., data with autocorrelation/cross correlation.

Based on the aforementioned description of the data, the probability distribution can be expressed as

$$P(\mathbf{X}) = \frac{1}{\sqrt{2\pi}} ((\det(\boldsymbol{\Sigma}))^{-1/2}) \exp\left(-\frac{1}{2} \mathbf{X}^T \boldsymbol{\Sigma}^{-1} \mathbf{X}\right). \quad (1)$$

In (1), inverse of the covariance ($\boldsymbol{\Sigma}^{-1}$) is replaced by $\boldsymbol{\Theta}$ and logarithmic form of $P(\mathbf{X})$ is considered to have tractable

computation. Thus, the expression in the following form is obtained as:

$$\log(P(\mathbf{X})) = -\frac{1}{2}\log(2\pi) + \frac{1}{2}\log(\det(\Theta)) - \frac{1}{2}(\mathbf{X}^T \Theta \mathbf{X}). \quad (2)$$

Furthermore, (2) can be modified as

$$\log(P(\mathbf{X})) = -\frac{1}{2}\log(2\pi) + \frac{1}{2}\log(\det(\Theta)) - \frac{1}{2}\text{Tr}(\mathbf{S}\Theta) \quad (3)$$

where $\mathbf{S}$ is the sample covariance matrix computed from the data and Θ is the inverse covariance matrix to be estimated. In order to estimate the inverse of covariance matrix, the problem is presented in the ML estimation framework as

$$\hat{\Theta} = \arg \min_{\Theta \succeq 0} (-\log(\det(\Theta)) + \text{Tr}(\mathbf{S}\Theta)). \quad (4)$$

In general, ML estimate of the inverse covariance matrix will lead to a dense graphical model. Therefore, inferring the connectivity among the variables becomes a challenging task and it may also lead to spurious connections. In order to overcome this issue, a sparsity constraint on the inverse of covariance matrix is considered [16].

The idea of sparsity has been extensively used in areas, such as machine learning, image and signal processing, mathematics, and statistics [16], [25]. Given the data with a large number of variables, the introduction of sparsity constraints helps in explaining the data with a minimum number of variables. In general, a sparsity constraint can be induced through regularization of parameters with L_0 (Zero) norm [22], L_1 -norm (LASSO) [26], or a combination of L_1 and L_2 norms (elastic net, CLOT) [27], [28]. Among the regularization techniques, L_0 provides the most sparse solution followed by L_1 and a combination of both L_1 - and L_2 -norms. Therefore, a sparsity constraint in terms of L_0 -norm on the inverse of covariance matrix is considered in this work.

It is a well-known fact that L_0 -norm minimization is combinatorial and is NP-hard to solve. Therefore, the original L_0 problem for inverse covariance estimation is generally transformed to its convex counterpart by relaxing the constraint to L_1 -norm [14], [17], [29]. Another way of handling the original L_0 problem is to use greedy methods, which makes locally optimal choice at every stage with a goal to reach the global optimum [22], [30]. As the greedy methods promote better sparsity over convex relaxations, the former is used in this article. In the following, details that are essential to understand both the convex formulations and greedy methods will be explained by considering a general function $f: \mathbb{R}^n \rightarrow \mathbb{R}$.

1) *One Norm (L_1) Constrained Optimization Algorithm: Convex Formulation:* A convex equivalent solution to the problem of inverse covariance estimation given in (4) is obtained by incorporating L_1 constraint on the decision variable to the objective function as

$$\hat{\Theta} = \arg \min_{\Theta \succeq 0} \log(\det(\Theta)) - \text{Tr}(\mathbf{S}\Theta) + \alpha \|\Theta\|_1. \quad (5)$$

To solve the aforementioned problem, ADMM is most prominently used. In this section, an overview of ADMM method for a general convex function $f(\theta) := h(\theta) + g(\theta)$ is presented in brief and its extension for SIC estimation is also

Algorithm 1 SIC Estimation: ADMM (Convex Formulation)

- 1: $\Theta^{k+1} = \mathbf{Q}\tilde{\Theta}^k \mathbf{Q}^T$ ▷ Smooth function update
 - 2: $\mathbf{Z}^{k+1} = \left(1 - \frac{\alpha}{\rho} |(\Theta^{k+1} + \mathbf{U}^k)|^{-1}\right)$ ▷ Non-smooth function update
 - 3: $\mathbf{U}^{k+1} = \mathbf{U}^k + \Theta^{k+1} - \mathbf{Z}^{k+1}$ ▷ Dual update
-

detailed in the later parts. Consider a minimization problem of a general convex function $f(\cdot)$ as

$$\begin{aligned} \hat{\theta} &= \arg \min_{\theta} f(\theta) \\ &= \arg \min_{\theta} h(\theta) + g(\theta). \end{aligned}$$

The aforementioned problem can be solved using different approaches, such as first-order methods [17], BCA [14], and ADMM [18]. In BCA, convergence becomes nonmonotonic for nonsmooth function because of which the estimate converges to a suboptimal solution. In order to avoid this issue, a new variable ($\mathbf{z}$) for θ is introduced in function $g(\cdot)$ and the unconstrained optimization problem is converted into a constrained form as

$$\begin{aligned} \hat{\theta} &= \arg \min_{\theta} h(\theta) + g(\mathbf{z}) \\ \text{s.t } &(\theta - \mathbf{z} = 0). \end{aligned}$$

This formulation, often referred to as ADMM, has the advantage of decomposability from dual descent and convergence properties from the method of multipliers [29]. The aforementioned problem is then converted into unconstrained form by introducing the method of multipliers and the final steps of estimating the parameters in ADMM method are as follows:

$$\begin{aligned} \theta^{k+1} &= \arg \min_{\theta} J_p(\theta, \mathbf{z}^k, \mathbf{y}^k) \\ \mathbf{z}^{k+1} &= \arg \min_{\mathbf{z}} J_p(\theta^{k+1}, \mathbf{z}, \mathbf{y}^k) \\ \mathbf{y}^{k+1} &= \mathbf{y}^k + \rho(\theta^{k+1} - \mathbf{z}^{k+1}) \end{aligned}$$

where $\rho > 0$ and $J_p(\theta, \mathbf{z}, \mathbf{y}) = h(\theta) + g(\mathbf{z}) + \mathbf{y}^T(\theta - \mathbf{z}) + (\rho/2)\|\theta - \mathbf{z}\|^2$.

In order to extend this approach to the case of estimation of SIC matrix, i.e., (5), the cost function defined in (5) is first separated into smooth $[h(\Theta) = \log(\det(\Theta)) - \text{Tr}(\mathbf{S}\Theta)]$ and nonsmooth functions ($g(\mathbf{Z}) = \|\Theta\|_1$). Applying the final steps of estimating the parameters in ADMM method for inverse covariance estimation, an analytical expression for inverse covariance matrix as given in Algorithm 1 is obtained [29].

In Algorithm 1, Θ^k is the estimated inverse covariance, $\mathbf{Z}^k$ is the variable, which regularizes the estimate with L_1 -norm, $\mathbf{U}$ is the dual variable of Θ , and $\mathbf{Q}$ is the matrix with orthonormal vectors obtained from singular value decomposition of Θ . The resultant graph structure estimated with L_1 constraint might make the structure dense if the regularization factor (α) is not tuned well, which may again lead to spurious relations.

2) *Zero Norm (L_0) Constrained Optimization Algorithm: Greedy Methods:* In this section, a brief description of L_0 -norm minimization problem is discussed by considering a general nonlinear function $f(\theta)$. The sparsity constrained

Algorithm 2 GSS Algorithm

- **Initialization:** Choose $\theta_0 \in C_s$ (Set of all vectors with at most s nonzero elements in each vector)
- **General step:** $k = 0, 1, 2, \dots, M$
 - If $\|\theta^k\|_0 < s$, define e_i to be an indicator vector whose i^{th} element is one and rest all zero. Compute for every $i = 1, 2, \dots, N$

$$\delta_i \in \arg \min_{\delta \in \mathbb{R}} f(\theta^k + \delta e_i) \quad (6)$$

$$f_i = f(\theta^k + \delta_i e_i) \quad (7)$$

Let $(i_k \in \arg \min_{i=1, \dots, N} f_i)$. If $f_{i_k} < f(\theta^k)$, then set

$$\theta^{k+1} = \theta^k + \delta_{i_k} e_{i_k} \quad (8)$$

Otherwise, STOP.

- If $\|\theta^k\|_0 = s$, define $I_1(\theta)$ to be the set of all non-zero indices of θ and e_i, e_j to be indicator vectors whose i^{th} and j^{th} elements are one and rest all zeros respectively. For every $i \in I_1(\theta^k)$ and $j = 1, \dots, N$ compute

$$\delta_i^j \in \arg \min_{\delta \in \mathbb{R}} f(\theta^k - \theta_i^k e_i + \delta e_j) \quad (9)$$

$$f_i^j = f(\theta^k - \theta_i^k e_i + \delta_i^j e_j) \quad (10)$$

Let $(i_k, j_k) \in \arg \min \{f_i^j : i \in I_1(\theta_k), j = 1, \dots, N\}$. If $f_{i_k}^{j_k} < f(\theta^k)$, then set

$$\theta^{k+1} = \theta^k - \theta_{i_k}^k e_{i_k} + \delta_{i_k}^{j_k} e_{j_k} \quad (11)$$

Otherwise, STOP.

optimization problem can be represented as

$$\begin{aligned} \min_{\theta} f(\theta) \\ \text{s.t. } \|\theta\|_0 \leq s \end{aligned}$$

where the loss function $f(\theta)$ is assumed to be smooth and lower bounded, $\|\theta\|_0$ represents the L_0 -norm of the parameter vector, and s is an integer that controls the level of sparsity. Furthermore, it is also assumed that the loss function can be either convex or nonconvex. Based on the assumptions mentioned above, several optimization algorithms are tailored to handle such problems [22]. Some of the algorithms detailed in [22] are guaranteed to converge to a basic feasible (BF) vector and some are guaranteed to converge to a coordinate-wise minimum (CW-minimum). The definitions of BF-vector and CW-minimum are given in the Appendix for the sake of completeness. In the hierarchy of optimal solutions, any optimal point is a CW-minimum and any CW-minimum is a BF-vector. Insights on the optimality conditions and the convergence properties can be obtained from [22]. Therefore, in this article, the GSS algorithm, which is guaranteed to converge to a CW-minimum, is used to solve the L_0 -norm minimization problem, and the algorithmic sequence of GSS approach is detailed in Algorithm 2

In the GSS algorithm, the main idea is to perturb the parameter vector $\theta \in C_s$ sequentially in each direction and the parameters are updated by using (8) and (11) based on

the optimal δ . This routine is continued until an optimal θ for the specified level of sparsity s is obtained. The sequence of perturbation at every iteration is achieved using the indicator vector e_i , which denotes the vector whose i^{th} element is one and rest all zero. An optimal δ in (6) and (9) can be obtained by using the first-order methods, where the derivatives of the functions in (6) and (9) with respect to δ are equated to zero. Several search-based methods, such as simplex method or genetic algorithms, can also be incorporated to estimate the optimal δ . However, it is to be noted that if the objective function is convex, the first-order methods provide advantage over search-based methods in terms of computation time for convergence. Therefore, the first-order methods are generally preferred for convex objective functions.

Since greedy methods promote better sparsity over convex formulations, the former is used in the current work of estimation of inverse covariance matrix. Therefore, the problem of estimation of inverse covariance is formulated as

$$\begin{aligned} \hat{\Theta} = \arg \min_{\Theta \succeq 0} (-\log(\det(\Theta)) + \text{Tr}(\mathbf{S}\Theta)) \\ \text{s.t. } \|\Theta\|_0 \leq s. \end{aligned} \quad (12)$$

Further details on solving the problem in (12) are discussed in Section III. Once the graph structure is estimated, the next objective is to infer the direction in which the variables are related, which is discussed in Section II-C.

C. Inferring Causal Direction

Given the inverse covariance matrix, causal direction is determined by computing the likelihood score, which is a metric that is used to compute the causal direction among the variables in an undirected graph. To explain further details on inference of causal direction using the likelihood score, two random variables x_i and x_j are considered. Given the functions $\hat{f}, \hat{g} : \mathbb{R} \rightarrow \mathbb{R}$ that are prediction of x_j from x_i and prediction of x_i from x_j , respectively, let the prediction residuals be defined as $\eta_{x_j, \hat{f}} := x_j - \hat{f}(x_i)$ and $\eta_{x_i, \hat{g}} := x_i - \hat{g}(x_j)$.

To infer causal relation among continuous random variables, differential entropy [denoted by $H(\cdot)$] is used [19]. For a continuous random variable $x \in \mathbb{R}$, differential entropy can be defined as $H(x) = \mathbb{E}[-\ln P(x)]$. Similarly, joint entropy can be represented as the sum of conditional differential entropy and marginal of the joint differential entropy using the chain rule of differential entropy [31] as

$$\begin{aligned} H(x_i, x_j) &= H(x_j) + H(x_i | x_j) \\ &= H(x_j) + H(\eta_{x_i, \hat{g}} | x_j) \\ &= H(x_j) + H(\eta_{x_i, \hat{g}}) - I(x_j, \eta_{x_i, \hat{g}}) \end{aligned}$$

where $I(\cdot, \cdot)$ represents the differential mutual information. For instance, given two random variables x_1 and x_2 , $I(x_1, x_2) := \mathbb{E}[\ln(p(x_1 | x_2)/p(x_2))]$, where $p(\cdot)$ represents the probability density function.

Similarly,

$$\begin{aligned} H(x_i, x_j) &= H(x_i) + H(x_j | x_i) \\ &= H(x_i) + H(\eta_{x_j, \hat{f}} | x_i) \\ &= H(x_i) + H(\eta_{x_j, \hat{f}}) - I(x_i, \eta_{x_j, \hat{f}}). \end{aligned}$$

The aforementioned equations of differential entropy lead to

$$\begin{aligned} H(x_i) + H(\boldsymbol{\eta}_{x_j, \hat{f}}) - I(x_i, \boldsymbol{\eta}_{x_j, \hat{f}}) \\ = H(x_j) + H(\boldsymbol{\eta}_{x_i, \hat{g}}) - I(x_j, \boldsymbol{\eta}_{x_i, \hat{g}}). \end{aligned} \quad (13)$$

In the case that the random variables x_i and x_j either satisfies an additive noise model with $x_i \rightarrow x_j$ or $x_j \rightarrow x_i$, but not both, then the mutual information of the corresponding term is zero [19], i.e., when the causal structure is given to be $x_i \rightarrow x_j$, then $I(x_i, \boldsymbol{\eta}_{x_j, \hat{f}}) = 0$ and $I(x_j, \boldsymbol{\eta}_{x_i, \hat{g}}) > 0$ and vice versa. Therefore, the differential entropy given in (13), for the causal structure $x_i \rightarrow x_j$ can be written as

$$\begin{aligned} H(\boldsymbol{\eta}_{x_j, \hat{f}}) + H(x_i) \\ = H(\boldsymbol{\eta}_{x_i, \hat{g}}) + H(x_j) - I(x_j, \boldsymbol{\eta}_{x_i, \hat{g}}) \\ \implies H(\boldsymbol{\eta}_{x_j, \hat{f}}) + H(x_i) \leq H(\boldsymbol{\eta}_{x_i, \hat{g}}) + H(x_j). \end{aligned} \quad (14)$$

Furthermore, when the random variables are assumed to be Gaussian, differential entropy can be related to variance and the following expression is obtained [19]:

$$\log \text{Var}(\boldsymbol{\eta}_{x_j, \hat{f}}) + \log \text{Var}(x_i) \leq \log \text{Var}(\boldsymbol{\eta}_{x_i, \hat{g}}) + \log \text{Var}(x_j). \quad (15)$$

Therefore, the score (often referred to as likelihood score) for the causal structure $x_i \rightarrow x_j$ is defined to be

$$L_{x_i \rightarrow x_j} : \log \text{Var}(\boldsymbol{\eta}_{x_j, \hat{f}}) + \log \text{Var}(x_i).$$

From (15), it can be noted that $L_{x_i \rightarrow x_j} < L_{x_j \rightarrow x_i}$ for the case when x_i causes x_j and $L_{x_j \rightarrow x_i} < L_{x_i \rightarrow x_j}$ when x_j causes x_i .

In this article, the aforementioned likelihood score is applied to the case when the function $\hat{f}(\cdot)$ is considered to be in the form of linear finite impulse response (FIR) models. Since negative log likelihood is most commonly used to infer the estimate of the parameters from the data for dynamic models, the likelihood score i.e., $L_{x_i \rightarrow x_j}^d$, can be defined in the context of differential entropy as

$$L_{x_i \rightarrow x_j}^d : -\log(\hat{\Sigma}_{(x_j)_t | (x_i)_{t-K+1}}) - \log(\hat{\Sigma}_{x_i}) \quad (16)$$

where $\hat{\Sigma}_{(x_j)_t | (x_i)_{t-K+1}} = \text{Var}(\boldsymbol{\eta}_{x_j, \hat{h}})$, $\boldsymbol{\eta}_{x_j, \hat{h}} = x_j(t) - \hat{h}(x_i(t), \dots, x_i(t-K+1))$, and $\hat{h}(\cdot)$ represents the linear FIR model. Since the undirected graph structure is known from the first step of the proposed method, the likelihood scores are computed by identifying an FIR model for the connected components (in a pairwise manner) of the estimated undirected graph. The likelihood score of a dynamic model is expected to be $L_{x_i \rightarrow x_j}^d > L_{x_j \rightarrow x_i}^d$ for the case when x_i causes x_j and $L_{x_j \rightarrow x_i}^d > L_{x_i \rightarrow x_j}^d$ when x_j causes x_i . Hence, this information can be used to infer the direction in the estimated undirected graph, thereby transforming undirected graph into a directed graph. It has to be noted that this score for computing the causal direction is only applicable if the variables are assumed to be Gaussian. For more details about the likelihood score, refer to [10] and [19].

III. SIC ESTIMATION USING GSS

In this section, the GSS approach is utilized to solve the inverse covariance estimation problem with zero-norm constraint, i.e., (12) along with positive semidefinite constraint for estimating the sparse graph structure. In particular, the expressions for minimization steps in the GSS algorithm, i.e., (6) and (9) when the function $f(\cdot)$ is expressed using (12), are detailed in this section. Furthermore, gradient-based methods are used to perform the minimization steps of (6) and (9) due to the convex nature of the objective function $f(\cdot)$. The main difference between the general structure of GSS approach and the method of minimization used in this article is that the optimization variable in this approach is a symmetric matrix. Therefore, a structured indicator matrix $\mathbf{E}$, instead of an indicator vector $\mathbf{e}$, is used to iterate through the elements of the matrix in an orderly sequence. The final steps in the GSS algorithm are also modified to account for the constraint of positive semidefiniteness of the solution matrix.

Considering (6), when the function $f(\cdot)$ is expressed using (12), the minimization problem reduces to

$$\begin{aligned} \delta_{i,j} &= \arg \min_{\delta \in \mathbb{R}} f(\boldsymbol{\Theta}^k + \delta \mathbf{E}_{i,j}) \\ &= \arg \min_{\delta \in \mathbb{R}} -\log(\det(\boldsymbol{\Theta}^k + \delta \mathbf{E}_{i,j})) + \text{Tr}(\mathbf{S}(\boldsymbol{\Theta}^k + \delta \mathbf{E}_{i,j})) \end{aligned} \quad (17)$$

where $i = 1, 2, \dots, M$, $j = i, i+1, \dots, M$, and $\mathbf{E}_{i,j}$ is a symmetric matrix with one at (i, j) th position and zeros elsewhere. Since gradient-based methods are used to solve the minimization problem, it is essential to compute the derivative of (17) with respect to δ . As the derivative computation is independent of indices i and j , for simplicity in notation, $\mathbf{E}_{i,j}$ is replaced with $\mathbf{E}$ in the following derivation and the indices are specified wherever required. Therefore,

$$-\text{Tr}\left[(\boldsymbol{\Theta}^k + \delta \mathbf{E})^{-1} \mathbf{E}\right] + \text{Tr}(\mathbf{S}\mathbf{E}) = 0. \quad (18)$$

Based on the inversion lemma for sum of matrices [32], $(\boldsymbol{\Theta}^k + \delta \mathbf{E})^{-1}$ in (18) can be given as follows:

$$(\boldsymbol{\Theta}^k + \delta \mathbf{E})^{-1} = (\boldsymbol{\Theta}^k)^{-1} - \frac{\delta}{1 + \delta \text{Tr}(\mathbf{E}(\boldsymbol{\Theta}^k)^{-1})} (\boldsymbol{\Theta}^k)^{-1} \mathbf{E} (\boldsymbol{\Theta}^k)^{-1}. \quad (19)$$

In (19), $(\delta)/(1 + \delta \text{Tr}(\mathbf{E}(\boldsymbol{\Theta}^k)^{-1}))$ is denoted as β . On substituting the inverse expression in (18) and after rearranging the terms, an expression of the following form can be obtained:

$$\beta \text{Tr}\left((\boldsymbol{\Theta}^k)^{-1} \mathbf{E}\right)^2 = \text{Tr}((\boldsymbol{\Theta}^k)^{-1} \mathbf{E}) - \text{Tr}(\mathbf{S}\mathbf{E}). \quad (20)$$

Using the symmetric polynomial property [33], for any square matrix $\mathbf{A}$, $\text{Tr}(\mathbf{A}^2) = [\text{Tr}(\mathbf{A})]^2 - 2 \sum_{i < j} \lambda_i \lambda_j$, where λ denotes the eigenvalue. Furthermore, when the rank($\mathbf{A}$) is one and $\mathbf{A}_{i,i} \neq 0$, then $\text{Tr}(\mathbf{A}^2) = [\text{Tr}(\mathbf{A})]^2$.

In the current problem, $((\boldsymbol{\Theta}^k)^{-1} \mathbf{E})$ has the rank of one when $\mathbf{E}$ operates on the diagonal element ($i = j$) and it has the rank of two when $\mathbf{E}$ operates on off-diagonal elements ($i \neq j$) due

to symmetric nature of $(\Theta^k)^{-1}$. Therefore,

$$\begin{aligned} & \text{Tr}\left[\left((\Theta^k)^{-1}\mathbf{E}\right)^2\right] \\ &= \begin{cases} \left[\text{Tr}\left((\Theta^k)^{-1}\mathbf{E}\right)\right]^2, & \text{for } i = j \\ \left[\text{Tr}\left((\Theta^k)^{-1}\mathbf{E}\right)\right]^2 - 2 \sum_{i < j} \lambda_i \lambda_j, & \text{for } i \neq j. \end{cases} \end{aligned} \quad (21)$$

Furthermore, denote $\text{Tr}[(\Theta^k)^{-1}\mathbf{E}]$ as γ and $\text{Tr}(\mathbf{SE})$ as κ . Then, for the case when $i \neq j$, (20) can be simplified as

$$\delta = \frac{\gamma - \kappa}{\gamma \kappa - 2 \sum_{i < j} \lambda_i \lambda_j}. \quad (22)$$

Note that when $i = j$, $\delta = (1/\kappa) - (1/\gamma)$.

Since the function is convex, the minimum of (6) is unique and is given by (22). Now, the function in (7) can be estimated as

$$\begin{aligned} & f_{i,j}(\Theta^k + \delta \mathbf{E}_{i,j}) \\ &= -\log \left[\det \left(\Theta^k + \left(\frac{\gamma - \kappa}{\gamma \kappa - 2 \sum_{i < j} \lambda_i \lambda_j} \right) \mathbf{E}_{i,j} \right) \right] \\ &+ \text{Tr}(\mathbf{S}\Theta^k) + \left(\frac{\gamma - \kappa}{\gamma \kappa - 2 \sum_{i < j} \lambda_i \lambda_j} \right) \kappa \end{aligned} \quad (23)$$

$(\hat{i}, \hat{j}) = \arg \min \{f_{i,j}, i = 1, 2, \dots, M, j = i, i + 1, \dots, M\}$ and the inverse covariance matrix can be updated as

$$\Theta^{k+1} = \begin{cases} \Theta^k + \left(\frac{\gamma - \kappa}{\gamma \kappa - 2 \sum_{i < j} \lambda_i \lambda_j} \right) \hat{\mathbf{E}}, & \text{when } \hat{i} \neq \hat{j} \\ [0.8pc]\Theta^k + \left(\frac{1}{\kappa} - \frac{1}{\gamma} \right) \hat{\mathbf{E}}, & \text{when } \hat{i} = \hat{j} \end{cases} \quad (24)$$

where $\hat{\mathbf{E}}$ is a symmetric matrix with one at the $(\hat{i}, \hat{j})$ th position and zeros elsewhere.

To ensure the positive semidefiniteness of inverse covariance matrix, the update will be modified [34] as

$$\begin{aligned} & [\mathbf{U}_{k+1}, \mathbf{D}_{k+1}, \mathbf{V}_{k+1}] = \text{svd}(\Theta^{k+1}) \\ & \hat{\Theta}^{k+1} = \mathbf{U}_{k+1} \max(\mathbf{D}_{k+1}, 0) \mathbf{V}_{k+1}. \end{aligned} \quad (25)$$

Considering (9), when the function $f(\cdot)$ is expressed using (12), the minimization problem reduces to

$$\begin{aligned} & \delta_{i_1, j_1}^{i, j} = \arg \min_{\delta \in \mathbb{R}} f(\Theta^k - \Theta_{i,j}^k \mathbf{E}_{i,j} + \delta \mathbf{E}_{i_1, j_1}) \\ &= \arg \min_{\delta \in \mathbb{R}} -\log(\det(\Theta^k - \Theta_{i,j}^k \mathbf{E}_{i,j} + \delta \mathbf{E}_{i_1, j_1})) \\ &+ \text{Tr}(\mathbf{S}\Theta^k) - \text{Tr}(\Theta_{i,j}^k \mathbf{SE}_{i,j}) + \delta \text{Tr}(\mathbf{SE}_{i_1, j_1}) \end{aligned} \quad (26)$$

where $i_1 = 1, 2, \dots, M$, $j_1 = i_1, i_1 + 1, \dots, M$, $(i, j) \in I(\Theta^k)$ with $I(\Theta^k)$ being a set of all nonzero indices of the matrix Θ^k , and $\mathbf{E}_{i,j}$ is a symmetric matrix with one at (i, j) th position and zero elsewhere. For simplicity in notation, $\mathbf{E}_{i,j}$ is represented as $\mathbf{E}_1$ and $\mathbf{E}_{i_1, j_1}$ is represented as $\mathbf{E}_2$ in the following derivation. Equating the derivative of (26) with respect to δ to zero results in

$$\text{Tr}\left[(\Theta^k + (\delta \mathbf{E}_2 - \Theta_{i,j}^k \mathbf{E}_1))^{-1} \mathbf{E}_2\right] - \text{Tr}(\mathbf{SE}_2) = 0. \quad (27)$$

Using the inversion lemma for sum of matrices [32], inverse portion in (27) can be replaced by

$$\begin{aligned} & \text{Tr}\left[\left((\Theta^k)^{-1} - \frac{(\Theta^k)^{-1}(\delta \mathbf{E}_2 - \Theta_{i,j}^k \mathbf{E}_1)(\Theta^k)^{-1}}{1 + \text{Tr}((\delta \mathbf{E}_2 - \Theta_{i,j}^k \mathbf{E}_1)(\Theta^k)^{-1})}\right) \mathbf{E}_2\right] \\ & - \text{Tr}(\mathbf{SE}_2) = 0. \end{aligned} \quad (28)$$

Denoting the term $(1)/(1 + \text{Tr}((\delta \mathbf{E}_2 - \Theta_{i,j}^k \mathbf{E}_1)(\Theta^k)^{-1}))$ in (28) as β^* , an expression of the following form is obtained after rearranging the terms as:

$$\begin{aligned} & \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_2 + (-\beta^*)(\Theta^k)^{-1}(\delta \mathbf{E}_2 - \Theta_{i,j}^k \mathbf{E}_1)(\Theta^k)^{-1} \mathbf{E}_2\right] \\ & - \text{Tr}(\mathbf{SE}_2) = 0. \end{aligned} \quad (29)$$

Based on the properties of trace, the aforementioned expression reduces to

$$\begin{aligned} & -\text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_2\right] + \beta^* \delta \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_2 (\Theta^k)^{-1} \mathbf{E}_2\right] \\ & - \beta^* \Theta_{i,j}^k \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_1 (\Theta^k)^{-1} \mathbf{E}_2\right] + \text{Tr}(\mathbf{SE}_2) = 0. \end{aligned} \quad (30)$$

To further simplify the notation, the trace terms in (30) are replaced using the following expressions:

$$\begin{aligned} & \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_1\right] = \gamma_1; \quad \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_2\right] = \gamma_2 \\ & \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_1 (\Theta^k)^{-1} \mathbf{E}_2\right] = \gamma_{12}; \quad \text{Tr}\left[(\Theta^k)^{-1} \mathbf{E}_2 (\Theta^k)^{-1} \mathbf{E}_2\right] = \gamma_{22} \\ & \text{Tr}(\mathbf{SE}_1) = \kappa_1; \quad \text{Tr}(\mathbf{SE}_2) = \kappa_2. \end{aligned}$$

Thus, (30) can be simplified and represented as follows:

$$\gamma_2 - \kappa_2 + \frac{\Theta_{i,j}^k \gamma_{12}}{1 + \delta \gamma_2 - \Theta_{i,j}^k \gamma_1} = \frac{\delta \gamma_{22}}{1 + \delta \gamma_2 - \Theta_{i,j}^k \gamma_1}. \quad (31)$$

Now, using the symmetric polynomial property, γ_{22} can be written as $[(\gamma_2)^2 - 2 \sum_{i < j} \lambda_i \lambda_j]$, where λ_i and λ_j are the eigenvalues of $[(\Theta^k)^{-1} \mathbf{E}_2]$. On substituting the equivalent expression for γ_{22} in (31) and after simple algebraic manipulations, the following expression can be obtained:

$$\Theta_{i,j}^k (\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2) + \gamma_2 - \kappa_2 - \delta \kappa_2 \gamma_2 = -2\delta \sum_{i < j} \lambda_i \lambda_j. \quad (32)$$

Then, by rearranging (32), the optimal expression for δ , when $i \neq j$, can be expressed as follows:

$$\delta = \frac{\gamma_2 - \kappa_2 + \Theta_{i,j}^k (\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2)}{(\kappa_2 \gamma_2 - 2 \sum_{i < j} \lambda_i \lambda_j)}. \quad (33)$$

When $i = j$, (33) modifies as follows:

$$\delta = \frac{1}{\kappa_2} - \frac{1}{\gamma_2} + \Theta_{i,j}^k \left(\frac{\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2}{\kappa_2 \gamma_2} \right) \quad (34)$$

and the function in (10) is given in (35), as shown at the bottom of the next page.

Now, $\{\hat{i}, \hat{j}, \hat{i}_1, \hat{j}_1\} = \arg \min \{f_{i_1, j_1}^{i, j}; i_1 = 1, 2, \dots, M, j_1 = i_1, i_1 + 1, \dots, M, (i, j) \in I(\Theta^k)\}$ and the update expression

Algorithm 3 SIC Estimation Using GSS Algorithm

- **Initialization:** Choose $\Theta_0 \in C_s$. Choosing an identity matrix or using solution from ADMM as initial guess would be an efficient choice.
- **General step:** For, $k = 0, 1, 2, \dots, M$
 - If $\|\Theta\|_0 < s$
 - * Estimate $(\hat{i}, \hat{j}) = \arg \min \{f_{i,j}, i = 1, 2, \dots, M, j = i, i+1, \dots, M\}$ where $f_{i,j}$ is defined in Eq. (23).
 - * If $f_{\hat{i},\hat{j}} < f(\Theta^k)$, update the covariance matrix using Eq. (25) and otherwise, stop the iteration.
 - If $\|\Theta\|_0 = s$,
 - * Estimate $\{\hat{i}, \hat{j}, \hat{i}_1, \hat{j}_1\} = \arg \min \{f_{i_1,j_1}^{i,j}; i_1 = 1, 2, \dots, M, j_1 = i_1+1, \dots, M, (i,j) \in I(\Theta^k)\}$ where, $f_{i_1,j_1}^{i,j}$ is defined using Eq. (35)
 - * If $f_{\hat{i}_1,\hat{j}_1}^{i,j} < f(\Theta^k)$, update parameter vector using Eq. (37); otherwise, stop the iteration.

for $\|\Theta^k\|_0 = s$ can be presented as follows:

$$\Theta^{k+1} = \begin{cases} \Theta^k - \Theta_{i,j}^k \hat{\mathbf{E}}_1 + \left(\frac{\gamma_2 - \kappa_2 + \Theta_{i,j}^k (\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2)}{(\kappa_2 \gamma_2 - 2 \sum_{i < j} \lambda_i \lambda_j)} \right) \hat{\mathbf{E}}_2 & \text{when } \hat{i} \neq \hat{j} \\ \Theta^k - \Theta_{i,j}^k \hat{\mathbf{E}}_1 + \left(\frac{1}{\kappa_2} - \frac{1}{\gamma_2} + \Theta_{i,j}^k \left(\frac{\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2}{\kappa_2 \gamma_2} \right) \right) \hat{\mathbf{E}}_2 & \text{when } \hat{i} = \hat{j} \end{cases} \quad (36)$$

where $\hat{\mathbf{E}}_1$ and $\hat{\mathbf{E}}_2$ are symmetric matrices with one at the $(\hat{i}, \hat{j})$ th and $(\hat{i}_1, \hat{j}_1)$ th position, respectively, and zeros elsewhere.

To ensure the positive semidefiniteness of inverse covariance matrix, the update equation will be modified [34] as

$$[\mathbf{U}_{k+1}, \mathbf{D}_{k+1}, \mathbf{V}_{k+1}] = \text{svd}(\Theta^{k+1}) \\ \hat{\Theta}^{k+1} = \mathbf{U}_{k+1} \max(\mathbf{D}_{k+1}, 0) \mathbf{V}_{k+1}. \quad (37)$$

The steps of GSS algorithm when the function $f(\cdot)$ is expressed using (12) are detailed in Algorithm 3.

Implementing the steps detailed in Algorithm 3, an undirected sparse graph/ network structure can be estimated from the data. Finally, to infer the causal direction among the variables in the estimated network, the likelihood score as detailed in Section II-C is utilized. The overall algorithm of the proposed method is given in Algorithm 4.

Algorithm 4 Causal Analysis With SIC

- 1) Estimate sparse inverse covariance matrix Θ using Algorithm 3.
- 2) Construct the undirected graph from Θ by connecting its non-zero components.
- 3) Compute pairwise likelihood score among the connected components in the graph using Eq. (16).
- 4) Accept the direction with maximum of computed likelihood score: $\max(L_{x_i \rightarrow x_j}^d, L_{x_j \rightarrow x_i}^d)$.
- 5) Repeat steps 3 & 4 for other pairs of variables.

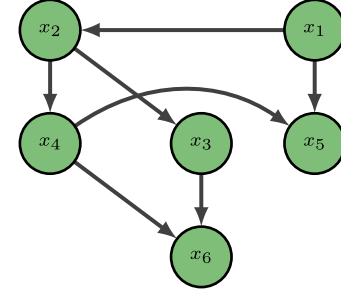


Fig. 2. Actual network model for numerical case study.

IV. CASE STUDIES

To demonstrate the efficacy of the proposed algorithm, two case studies are considered in this article. The first case study is a numerical example where a constructed system in which interactions among the variables are known *a priori* is considered. The second case study is an industrial problem, in which the objective is to construct a causal network for inferring the phenomena of tower flooding and weeping in a deethanizer column associated with FCC unit.

A. Numerical Example

In this case, data with six variables are considered for the study. A network model representing the actual causal relationship among the variables is shown in Fig. 2.

True inverse covariance or the precision matrix for the network in Fig. 2 is given as follows:

$$\Theta_{\text{true}} = \begin{bmatrix} 14.47 & 4.80 & 0 & 0 & 3.90 & 0 \\ 4.80 & 8.47 & 1.75 & 6.00 & 0 & 0 \\ 0 & 1.75 & 10.47 & 0 & 0 & 3.40 \\ 0 & 6.00 & 0 & 11.47 & 3.80 & 1.90 \\ 3.90 & 0 & 0 & 3.80 & 7.45 & 0 \\ 0 & 0 & 3.40 & 1.90 & 0 & 12.42 \end{bmatrix}.$$

$$f_{i_1,j_1}^{i,j}(\Theta^k - \Theta_{i,j}^k \mathbf{E}_1 + \delta \mathbf{E}_2) = -\log \left(\det \left(\Theta^k - \Theta_{i,j}^k \mathbf{E}_1 + \left(\frac{\gamma_2 - \kappa_2 + \Theta_{i,j}^k (\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2)}{(\kappa_2 \gamma_2 - 2 \sum_{i < j} \lambda_i \lambda_j)} \right) \mathbf{E}_2 \right) \right) \\ + \text{Tr}(\mathbf{S} \Theta^k) - \text{Tr}(\Theta_{i,j}^k \mathbf{S} \mathbf{E}_1) + \left(\frac{\gamma_2 - \kappa_2 + \Theta_{i,j}^k (\gamma_{12} - \gamma_1 \gamma_2 + \gamma_1 \kappa_2)}{(\kappa_2 \gamma_2 - 2 \sum_{i < j} \lambda_i \lambda_j)} \right) \text{Tr}(\mathbf{S} \mathbf{E}_2) \quad (35)$$

TABLE I
LIKELIHOOD SCORES FOR NUMERICAL CASE STUDY

Variable Pairs		$L_{x_i \rightarrow x_j}^d$	$L_{x_j \rightarrow x_i}^d$	max (Score)
x_1	x_2	1.93	1.43	1.93
x_1	x_5	2.00	1.39	2.00
x_2	x_3	0.46	0.33	0.46
x_2	x_4	0.33	0.13	0.33
x_4	x_5	3.01	2.66	3.01
x_3	x_6	0.19	0.16	0.19
x_4	x_6	0.32	0.30	0.32

Considering the inverse covariance matrix given in Θ_{true} as a reference and with zero mean, 1000 samples of data for each node are generated. Using the computed covariance ($\mathbf{S}$) from the generated data, SIC is estimated from the convex (ADMM) and nonconvex (proposed) methods for the reconstruction of undirected network structure. Estimates of the inverse covariance matrix obtained using ADMM after thresholding and the proposed GSS method are as follows:

$$\hat{\Theta}_{\text{ADMM}} = \begin{bmatrix} 13.51 & 4.35 & 0 & -0.21 & 3.53 & 0 \\ 4.35 & 7.39 & 1.46 & 5.49 & -0.19 & 0 \\ 0 & 1.46 & 10.04 & 0 & 0 & 3.02 \\ -0.21 & 5.49 & 0 & 10.71 & 3.49 & 1.71 \\ 3.53 & -0.19 & 0 & 3.49 & 7.12 & 0 \\ 0 & 0 & 3.02 & 1.71 & 0 & 11.90 \end{bmatrix}$$

$$\hat{\Theta}_{\text{GSS}} = \begin{bmatrix} 14.20 & 4.82 & 0 & 0 & 3.76 & 0 \\ 4.82 & 8.16 & 1.69 & 5.71 & 0 & 0 \\ 0 & 1.69 & 10.46 & 0 & 0 & 3.51 \\ 0 & 5.71 & 0 & 11.08 & 3.62 & 1.79 \\ 3.76 & 0 & 0 & 3.62 & 7.28 & 0 \\ 0 & 0 & 3.51 & 1.79 & 0 & 12.18 \end{bmatrix}.$$

By connecting the nonzero components of the estimated matrix, it can be noted that the proposed method is able to accurately reconstruct the true underlying structure. On the other hand, spurious connections are present in the estimate obtained using the ADMM method. Furthermore, to quantify the accuracy of the estimate, the Frobenius norm ($\|\cdot\|_F$, defined as the square root of sum of squares of all the elements in the matrix) is computed between $\hat{\Theta}$ and Θ_{true} . For the numerical case study, a value of 0.937 is obtained for GSS and a value of 2.03 for ADMM is obtained. From these results, it can be concluded that the proposed approach provides a relatively more accurate estimate compared to its convex counterpart. Also, the number of nonzero entries in the estimate obtained using the proposed method is less than that of the estimate obtained using ADMM, from which it can be concluded that the proposed method is promoting better sparsity.

To infer the causal direction, likelihood score is computed among the variables in the undirected network that is estimated using the proposed method. Scores computed among the connected nodes in the estimated network are given in Table I.

Using the estimated undirected network structure and the scores computed, directed network that explains the causal relation is constructed and is given in Fig. 3.

In addition to the aforementioned case study, 1000 samples of autocorrelated and/or cross-correlated data are generated using a discrete-time linear dynamic model presented in the

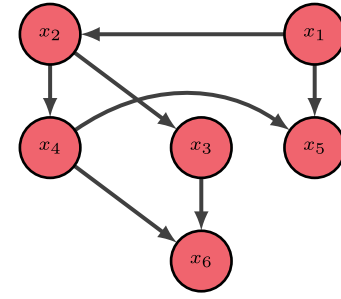


Fig. 3. Reconstructed causal network using the proposed approach.

following equation:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{x}(t-1) + \boldsymbol{\epsilon}(t) \quad (38)$$

where $\mathbf{x}(t) = [x_1(t), \dots, x_6(t)]^T$ is a variable vector, $\boldsymbol{\epsilon}(t) = [\epsilon_1(t), \dots, \epsilon_6(t)]^T$ is an error vector following Gaussian distribution with a zero mean and covariance $[\Theta_{\text{true}}]^{-1}$, and $\mathbf{A}$ is a square matrix encoding the model parameters presented as

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}.$$

Based on the data generated using a linear dynamic model, the SIC matrix is estimated using both the ADMM approach and the proposed algorithm. The estimates are obtained as follows:

$$\hat{\Theta}_{\text{ADMM}}^d = \begin{bmatrix} 11.14 & 3.11 & 0 & -0.64 & 2.41 & 0 \\ 3.11 & 7.17 & 1.43 & 4.79 & -0.58 & -0.13 \\ 0 & 1.43 & 8.66 & -0.09 & 0 & 1.86 \\ -0.64 & 4.79 & -0.09 & 9.73 & 2.82 & 1.65 \\ 2.41 & -0.58 & 0 & 2.82 & 6.65 & 0 \\ 0 & -0.13 & 1.86 & 1.65 & 0 & 10.29 \end{bmatrix}$$

$$\hat{\Theta}_{\text{GSS}}^d = \begin{bmatrix} 12.52 & 3.76 & 0 & 0 & 3.44 & 1.04 \\ 3.76 & 7.88 & 1.39 & 5.04 & 0 & 0 \\ 0 & 1.39 & 9.54 & 0 & 0 & 2.12 \\ 0 & 5.04 & 0 & 9.92 & 3.11 & 1.62 \\ 3.44 & 0 & 0 & 3.11 & 7.04 & 0 \\ 1.04 & 0 & 2.12 & 1.62 & 0 & 11.48 \end{bmatrix}.$$

It can be noted that both the estimates, $\hat{\Theta}_{\text{GSS}}^d$ and $\hat{\Theta}_{\text{ADMM}}^d$, exhibit spurious connections. However, the estimate obtained using the proposed approach yields relatively less spurious connections than with the ADMM method ($(\|\hat{\Theta}_{\text{GSS}}^d\|_0 = 22) < (\|\hat{\Theta}_{\text{ADMM}}^d\|_0 = 28)$). Furthermore, to quantify the estimate's accuracy, the Frobenius norm between $\hat{\Theta}_{\text{GSS}}^d$ and Θ_{true} is computed. In the second numerical example, a value of 6.52 is obtained for GSS and a value of 6.84 is obtained for ADMM. Since $\hat{\Theta}_{\text{GSS}}^d$ has relatively less error and has less spurious connections than $\hat{\Theta}_{\text{ADMM}}^d$, it can be concluded that the proposed method is relatively more accurate. To infer the causal direction, the likelihood score is computed among the nonzero entries of the $\hat{\Theta}_{\text{ADMM}}^d$ matrix. Likelihood score for

TABLE II
LIKELIHOOD SCORES FOR NUMERICAL CASE STUDY WITH
CORRELATED DYNAMIC DATA

Variable Pairs		$L_{x_i \rightarrow x_j}^d$	$L_{x_j \rightarrow x_i}^d$	max (Score)
x_1	x_2	1.16	1.05	1.16
x_1	x_6	2.17	2.04	2.17
x_1	x_5	2.20	2.01	2.20
x_2	x_3	0.84	0.81	0.84
x_2	x_4	0.53	0.49	0.53
x_4	x_5	0.91	0.88	0.91
x_3	x_6	3.02	2.96	3.02
x_4	x_6	0.77	0.74	0.77

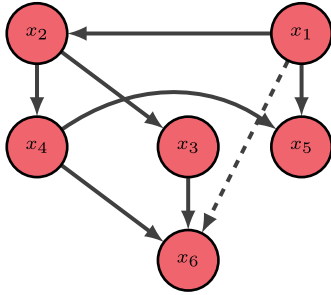


Fig. 4. Reconstructed causal network using the proposed approach for correlated dynamic data. Spurious connection (between x_1 and x_6) is indicated through dotted lines.

autocorrelated and cross-correlated data is presented in Table II along with the reconstructed causal network in Fig. 4.

B. Industrial Application: Tower Flooding and Weeping Problem

In this section, the proposed method is applied to an industrial case study, i.e., tower flooding and weeping problem. Flooding and weeping phenomena can occur in packed columns or tray towers (sieve trays/bubble columns) that are used for separation in process industries. Although flooding can occur in both packed column and tray towers, weeping will happen only in tray towers. These two phenomena will lead to poor separation efficiency; thereby, product quality is reduced and energy consumption is increased. In such a scenario, process upset and process throughput reductions are observed due to flooding and weeping. Occasionally, these phenomena may drive the entire process to shut down for days and weeks. Therefore, it is essential to analyze the probable reasons for occurrence of such phenomena so that the plant can be operated in a normal mode. In this work, causality analysis is carried out by using the proposed method, in order to identify the source for such problems in tray tower distillation column. In the current study, the flooding and weeping phenomena occurring in the deethanizer column associated with the FCC unit are considered and the results of the proposed method are validated using the underlying physics of the process as well as subsequent application of the proposed solution in the actual process.

1) *Process Description:* In this study, FCC and process units associated with it in the upstream and the downstream, such as feed preheating unit, fractionator unit, condenser, and deethanizer column, are considered. A schematic of the entire process is given in Fig. 6.

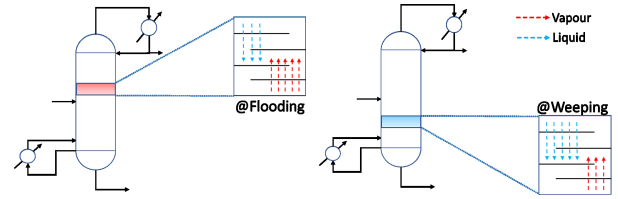


Fig. 5. Illustration of flooding and weeping.

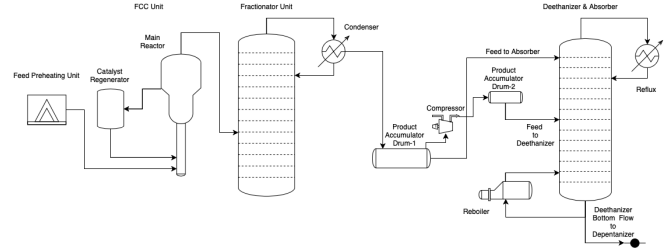


Fig. 6. Schematic of the industrial process.

In this process, the products from the hydrotreating units enter the feed preheating unit as feed. Once the preheating is complete, feed enters the FCC reactor where it undergoes cracking in the presence of catalyst. After cracking, the product enters the fractionator unit as feed in the gaseous phase. During the separation, multiple side cuts are drawn at different temperatures and finally top product from the separation enters the condenser. The products from the condenser enter the product accumulator drum-1 and then splits into two streams, where the product in one of the streams enters the absorber and the product in the other stream is compressed and enters the deethanizer via product accumulator drum-2 for further separation. The deethanizer unit has reboiler at the bottom and reflux on the top. In this process, reboiler is used as control handle to maintain the temperature of the trays at the bottom section of the deethanizer, and at the same time, the tray temperatures are used to control the phenomena of flooding. Finally, the products from the bottom of the deethanizer are processed for further separation.

Flooding is a phenomenon in which liquid flowing in the counter-current direction from the top of the distillation column gets entrained in the tray of a column due to the flow of vapor at high pressure from the bottom of the column. Similarly, weeping is the phenomenon that happens in a tray tower when the liquid starts flowing down through the sieve holes due to the lack of pressure exerted by the vapor flowing upward. Both these phenomena are shown in Fig. 5. Based on the process knowledge, a total of four hypotheses are expected to be the possible cause for the onset of either flooding or weeping in the deethanizer and these four possible hypotheses are listed in the following. It is to be noted that the existing method reported in the literature [35] utilizes the pressure drop information from the column to detect the onset of flooding. In the present case study, the pressure drop information is not available, and hence, flooding and the primary causes for flooding are identified through other source variables determined by causal analysis.

1) *Hypothesis-1:* First hypothesis is considered to be the change of feed flowing into the FCC unit. Whenever

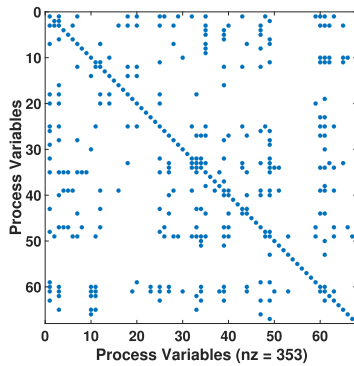


Fig. 7. SIC estimate using the proposed method.

there is a drastic feed change in FCC reactor, it leads to a reduction in flow of feed to the subsequent process units. Eventually, this reduces the feed for the deethanizer. When feed for the deethanizer reduces, preoccupied liquid in the bottom of the column vaporizes at a higher rate and this causes flooding.

- 2) *Hypothesis-2*: Second hypothesis is with respect to the abnormal temperature in reboiler associated with the deethanizer. When increase of the temperature of the reboiler is not related to the feed entering the deethanizer, the bottom tray temperature of the unit increases the flow of vapor inside the column. This excessive flow of vapor inside the column eventually causes flooding.
- 3) *Hypothesis-3*: Third hypothesis considers feed temperature of the deethanizer. When feed temperature increases at slower rate, energy balance of the column changes, thereby causing flooding.
- 4) *Hypothesis-4*: Fourth hypothesis is associated with weeping. Based on the knowledge obtained from process operation, it is understood that improper control of flooding leads to weeping due to sudden change in the operating condition of the deethanizer.

In the following, nature of the process data will be detailed.

2) *Process Data*: In order to estimate the SIC matrix, process data from the routine operation are considered. Before the estimation step, process data are normalized. The process data include 67 process variables, such as flow rates, pressure, and temperature associated with the process units (Fig. 6). They are sampled every minute, over the period of five months i.e., approximately 200 000 samples.

3) *Outcome*: Upon computing the sample covariance from process data, SIC is estimated and causal direction is computed using likelihood score. SIC estimated using GSS is given in Fig. 7 and the one estimated using ADMM is given in Fig. 8.

In the given estimate, nonzero components represented in blue are connected to reconstruct relationship among the variables. It can also be noted that the proposed method is promoting more sparsity than ADMM. Hence, the estimate obtained using the proposed method is used for further analysis.

Since the primary objective of this work is to infer the causal map for the flooding and weeping phenomena, process

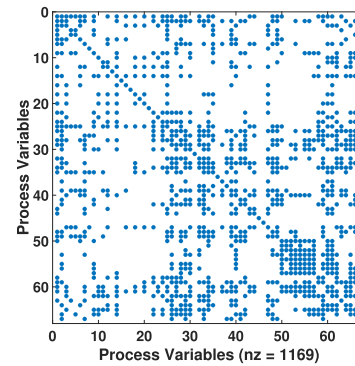


Fig. 8. ADMM estimate of SIC matrix.

TABLE III
PROCESS VARIABLES DESCRIPTION

Symbol	Process Variable Description
<i>ROT</i>	Reboiler Outlet Temperature
<i>FF</i>	Feed for FCC Reactor
<i>AF</i>	Feed for Absorber
<i>FT</i>	Temperature in FCC Reactor
<i>DIT</i>	Deethanizer Inlet Flow Temperature
<i>FI</i>	Flooding Indicator Variable (Flow variable)
<i>WI₁ & WI₂</i>	Weeping Indicator Variables (Deeth. Tray Temp. 1, 2)
<i>DIF</i>	Deethanizer Inlet Flow
<i>DBT</i>	Deethanizer Bottom Tray Temperature
<i>DFT</i>	Deethanizer Feed Temperature (From storage tank-2)
<i>CPC</i>	Wet Gas Compressor Pressure
<i>DFV</i>	Deethanizer Bottom Flow variable

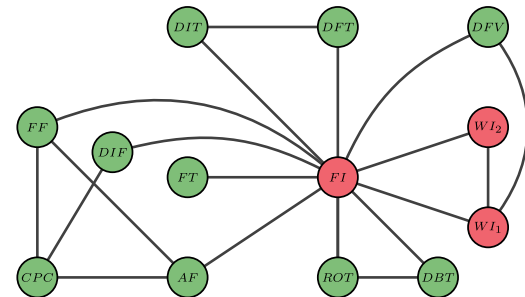


Fig. 9. Estimated undirected graph for flooding and weeping.

variables that are associated with these indicators are used in the analysis. In Fig. 7, the first row has the information of flooding indicator and variables that are associated with it, and similarly, the third and eighteenth rows have the information of weeping and variables associated with them. Also, the variables that are redundant are not considered for causal analysis and the physical interpretation of the remaining associated variables is given in Table III.

Undirected graph constructed among the subset of variables associated with flooding and weeping is shown in Fig. 9. In the graph, flooding and weeping indicators are shown in red and rest all in green.

Upon constructing the undirected graph, direction of cause and effect is inferred using likelihood score and the scores are given in Table IV.

Using the information of likelihood score, presented in Table IV, information of causal direction is incorporated into the undirected graph and the resulting causal network along with the hypothesis is presented in Fig. 10. To validate

TABLE IV
LIKELIHOOD SCORES FOR FLOODING AND WEEPING PROBLEM

Variable Pairs		$L_{x_i \rightarrow x_j}^d$	$L_{x_j \rightarrow x_i}^d$	$\max(\text{Score})$	Variable Pairs		$L_{x_i \rightarrow x_j}^d$	$L_{x_j \rightarrow x_i}^d$	$\max(\text{Score})$
x_i	x_j				x_i	x_j			
FF	FI	12.76	11.06	12.76	WI ₂	FI	10.78	12.96	12.96
FT	FI	13.91	11.50	13.91	FF	CPC	12.64	10.23	12.64
AF	FI	12.67	11.57	12.67	FF	AF	16.35	15.02	16.35
ROT	FI	16.12	12.60	16.12	CPC	AF	12.76	11.56	12.76
DBT	FI	15.88	12.97	15.88	WI ₁	WI ₂	10.40	16.79	16.79
DIT	FI	16.64	12.48	16.64	DIT	DFT	15.03	16.54	16.54
DFT	FI	14.20	11.33	14.20	DFV	WI ₁	14.49	16.45	16.45
DFV	FI	12.22	13.30	13.30	ROT	DBT	21.54	20.09	21.54
WI ₁	FI	12.52	14.52	14.52	CPC	DIF	11.44	13.05	13.05
DIF	FI	12.81	11.28	12.81					

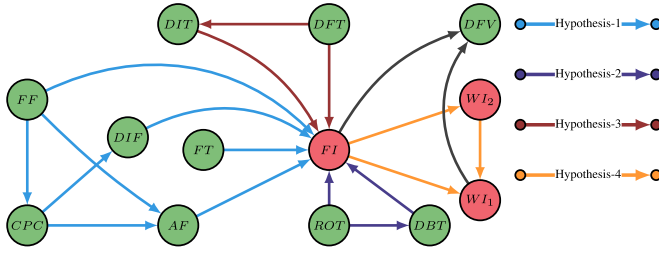


Fig. 10. Causal network for flooding and weeping.

the causal network, connections in the network are interpreted using process knowledge. In the causal network, four primary variables for flooding and one primary variable for weeping have been indicated as possible primary cause along with their relation with other process variables. In the case of flooding, the proposed approach indicates that changes in feed for FCC reactor (FF), deethanizer feed temperature (DFT), reboiler outlet temperature (ROT), and FCC reactor temperature (FT) have a direct effect on flooding (FI). Along with this, the effect of propagation occurring due to the change in the primary causal variables has been indicated through wet gas compressor pressure (CPC), absorber feed (AF), deethanizer inlet flow (DIF), deethanizer inlet flow temperature (DIT), and deethanizer bottom tray temperature (DBT). Individually, change in FCC reactor feed and CPC, causes AF and deethanizer feed to change thereby reducing the flow of liquid into the deethanizer. At the same time, change in ROT causes deethanizer bottom temperature to change, thereby increasing the temperature at the bottom section of the deethanizer. Similarly, change in FT and DFT will contribute to the change in temperature inside the deethanizer by increasing the temperature of the feed, which eventually leads to undesired operating condition. Considering the changes in FCC reactor feed and ROT simultaneously along with the changes in FCC feed temperature and DFT, it can be noticed that the outcomes achieved with the proposed method validate the reasons for flooding explained by hypothesis-1, hypothesis-2, and hypothesis-3. Similarly, for weeping, it can be noticed that flooding causes weeping due to the change in deethanizer tray temperatures (WI₁ and WI₂), and among the weeping indicators, deethanizer tray temperature-2 causes deethanizer tray temperature-1 to change. On comparing this outcome with the process knowledge explained in hypothesis-4, it is understood that control of flooding leads to change

in conditions inside the deethanizer thereby causing weeping. However, if the control of flooding is performed in a timely manner, with a good flooding predictor, weeping can be avoided. Among the variables indicating weeping, deethanizer tray temperature-2, located closer to flooding indicator and deethanizer tray temperature-1, is measured at the bottom of the column. Therefore, due to the system delay, deethanizer tray temperature-2 causes deethanizer tray temperature-1 to change. Then, at last, effects of flooding and weeping gets encoded in deethanizer bottom outlet flowrate (DFV). The end result of causal network estimation using the proposed method helped us to identify the reasons that are physically interpretable, for the problem of flooding and weeping in the deethanizer.

V. CONCLUSION

In this article, a two-step approach has been proposed for causal analysis. In the first step, a sparse undirected network is reconstructed from the SIC matrix with correlation information, and in the second step, the causal direction is determined using a likelihood score. In the proposed method, an L_0 -norm constraint is considered while estimating the SIC matrix and the resultant problem is solved using the GSS algorithm. The algorithm is suitably modified by incorporating positive semidefinite constraint on the decision variable. The efficiency of the proposed approach has been demonstrated using numerical examples and an industrial case study. In particular, the main objective of industrial case study is to infer causal relationships in order to address the problem of flooding and weeping in the deethanizer unit. From the results, it is observed that the proposed method is able to identify causal connections accurately for both numerical and industrial case studies. From the results of industrial case study, it can be further concluded that the probable reasons for flooding phenomena in a deethanizer unit are the drastic feed change for FCC reactor, DFT, and the reboiler temperature of deethanizer. Similarly, flooding is considered to be the probable reason for occurrence of weeping in the deethanizer unit.

APPENDIX

Letting the set of all vectors with at most s nonzero elements in each vector be denoted as C_s and $I(\theta)$ representing the indices of nonzero elements in the vector θ , the following definitions are given considering the minimization problem $\min_{\|\theta\|_0 \leq s} f(\theta)$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}$.

BF Vector: A vector, $\theta^* \in C_s$, is called a BF vector to the above problem if the following conditions hold.

- 1) $\|\theta^*\|_0 < s \implies \nabla f(\theta^*) = 0$.
- 2) $\|\theta^*\|_0 = s \implies \nabla_i f(\theta^*) = 0 \forall i \in I(\theta^*)$.

Coordinatewise Minimum: If θ^* is a feasible solution of the optimization problem, then it is called a coordinatewise minimum if the following conditions hold.

- 1) $\|\theta^*\|_0 < s \implies f(\theta^*) = \min_{\delta \in \mathbb{R}} f(\theta^* + \delta e_i) \forall i = 1, 2, \dots, n$.
- 2) $\|\theta^*\|_0 = s \implies f(\theta^*) \leq \min_{\delta \in \mathbb{R}} f(\theta^* - \theta_i^* e_i + \delta e_j) \forall i \in I(\theta^*), j = 1, 2, \dots, n$.

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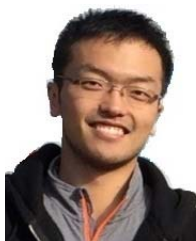
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